

IoT-Based Water-Leak Localisation and Selective Isolation

Technical Progress Report

Team IoNexus · IoTrix 2.0 Semi-Final · Track A: Embedded IoT System · 5 October 2026

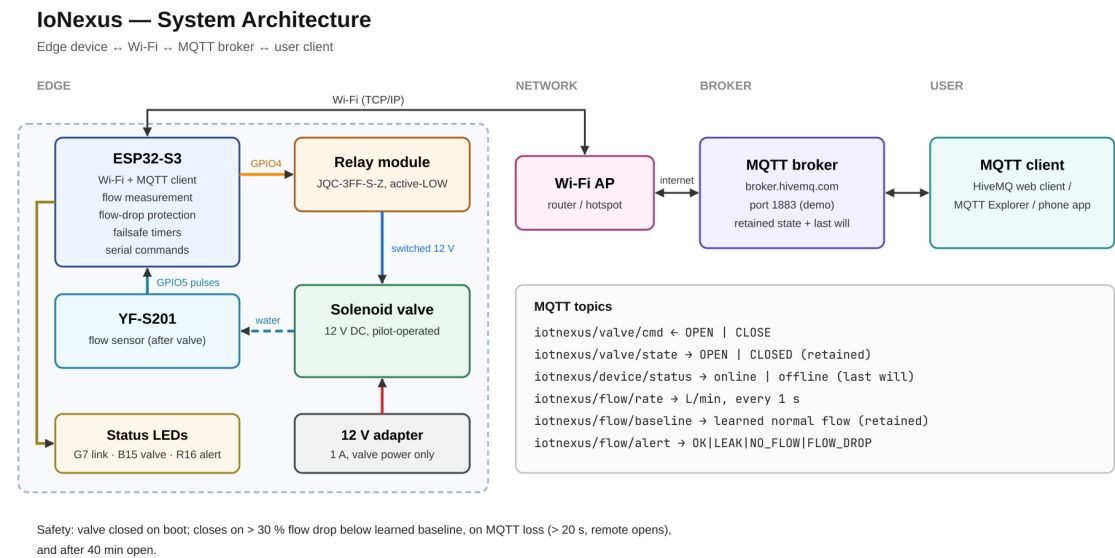
Problem

Leaks in water pipes go unnoticed until water is wasted or damage occurs. Their location is unknown, and the only control is usually the main valve, so stopping a leak cuts supply to every user.

Proposed Solution

The pipe network is divided into segments, each with an inlet solenoid valve, an outlet flow sensor and an ESP32 node. The node learns the segment's normal flow; a leak lowers the delivered flow, so a sustained drop ($>30\%$ for 3s) locates the leak to that segment and the node closes only its own valve. Other segments stay supplied. Alerts and live flow are sent over MQTT; all decisions run on the node.

Architecture



Current Progress

One complete segment node built: valve control through a relay, flow measurement, status LEDs, leak-detection state machine (settle → learn baseline → monitor), **FLOW_DROP/LEAK/NO_FLOW** alerts, failsafes (closed on boot, on network loss, after max open time), Wi-Fi/MQTT with auto-reconnect, and local serial control. Hardware validated: valve, relay, flow sensor, LEDs. Leak logic validated in simulation.

Technology Stack

ESP32-S3 DevKitC-1 · JQC-3FF-S-Z relay · 12 V pilot-operated solenoid valve · YF-S201 flow sensor · C++/Arduino on PlatformIO · PubSubClient · Wi-Fi 2.4 GHz · MQTT 3.1.1 (HiveMQ broker) · Git/GitHub

Testing

Valve debugging: pulsed leaking traced to the valve fitted against its flow arrow (fixed); valve needs a pressurised supply, tap pressure was insufficient. **Components**: LEDs, relay + valve and flow sensor passed; flow starts ≈ 7 s after opening, reaches 4.5–5.5 L/min, returns to zero within ≈ 3 s of closing. **Simulation**: with the firmware's logic, a simulated leak 30 s after opening (1.41→0.93 L/min) was detected and the valve closed 3 s later with a **FLOW_DROP** alert.

Limitations

Supply-pressure variation can look like a leak (main risk); valve needs minimum inlet pressure; leaks below the 30% threshold are missed; flow sensor not yet calibrated; prototype covers one segment; public broker without authentication; no flyback diode yet.

Next Step

Run the full system tests on hardware (leak isolation, no false trips, network loss, power cycle); calibrate the flow sensor; add an inlet flow sensor to compare inlet and outlet flow and reject pressure variation; build a second node to demonstrate selective isolation between segments.